



Comparing Categories of Plots Transcript

Welcome to BITS Pilani Digital. Today we will be continuing our learning journey of our course data visualisation and storytelling and we will be looking at taxonomy of data visualisation methods. In this lesson we will be trying to understand what are the different chart types or graphics and understand their general purpose.

In this context we will try to connect with earlier learnings that we had in this course where we have emphasised the need to choose the right graphic for the right purpose. Now we will be dwelling deep how to choose the best visual for the given purpose. Now we will be going deep understanding what are the right graphs for choosing comparisons or understanding hierarchies.

We will focus on the latest graphics that are available in the market or the softwares that are there. The focus is not on the traditional plots but on the advanced plots that are used for doing comparisons or assessing hierarchies. The choice is because to expose you to the latest development so that you can use them for building customised visuals for delivering better visualisations.

Now take a close look at this table. For comparing categories the main purpose is you have to the communication purpose for using comparison plots or comparing categories is to show how relative and absolute values of the variables are changing. For example you want to compare let us say marks scored by two students across six subjects.

It's a simple comparison and you can plot their marks scored across six subjects using a bar graph. This is a very simple comparison. The purpose here is to compare.

You can choose bar chart or you can also choose line chart but bar chart is most suited for this purpose. Now assessing part to whole relationships. This is done to understand how composition is varying across the constituent elements of a category.

This is very important to understand when you have a distribution within different categories and how they are changing over a period of time or across variables. For this you can use a stacked bar chart which is a very common thing to use or nowadays you have area charts and circle to path whole relationship the new graphs that help you to do this in a very intuitive manner and when they are very appealing for the audience. And now we can also showcase changes over time which are temporal data which we will be doing using different types of line graphs.

And finally you have plotting connections and relationships the network graphs and mapping geospatial data. The last three we will be taking in the next session. We will be now focussing on comparisons and hierarchy tips.

To comparing categories we will be now looking at a very interesting chart called Gantt chart.

The main purpose of this chart is to enable relative and absolute comparisons across categorical values. For example here we are seeing different countries in fact the variable of interest is not very important.

It shows that the particular variable has different ranges in different countries. Now the point here is if you were to show this in a bar graph it will become really difficult because certain countries will not have the values between 0 to 40 they are only between 40 to 80. It becomes really tough to show these values in a bar graph.

In a Gantt chart bar graph is allowed to move freely across the axis. This enables it to present the relative and absolute values easily. By looking at the graph we can simply say that Korea has relatively smaller values compared to Germany or France and all this information is contained in one graphic.

If the same bar graph by allowing flexibility to move around is enabled to do a comparison on the relative and the absolute scale on the simultaneous basis. This is the advantage of using a Gantt chart. Now we take another chart that enables comparing categories called Sankey chart.

For example you have the sources of energy, electricity, natural gas, coal and oil and you have the different types of customers consuming these energies residential, commercial, industrial and transport. Traditionally how would have drawn this plot? You would have had electricity consumption on a stacked bar chart electricity is being produced and it is being used for four different types of categories and share of their consumption of electricity from an electricity. Share of an in this example you can see the different sources of energy, electricity, coal, natural gas and oil and these are used by four categories of customers residential, commercial, industrial and transport and you can see the bars or the flows that are connecting shows the proportion of the energy used by that particular segment of the customers.

For example electricity is majorly used by the residential customers followed by the commercial category and almost nothing by industrial or transport. As opposed to this natural gas only a few percent is going to a residential majority is going towards industrial and it is going towards transport. Likewise coal is going towards commercial and a bit towards transport and you have oil which goes through industrial and transport for the most of the part.

In this way Sankey diagram helps you to see the flow between two categorical variables represented in the quantitative feature simultaneously capturing the categories. It also is used for highlighting the series of events that are there in between and associations that are there between the categories of variables. In this chart you can see the consumption of energy sources by the residential, commercial, industrial and transport sectors of the economy and their corresponding dependence on different energy sources like electricity, natural gas, oil and coal.

The width of the line or the band shows the nature of dependence the extent of

dependence of these categories and one can understand readily that residential customers are heavily dependent on electricity and so as the commercial customers. Alternatively you can see electricity is being majorly used by residential and commercial customers as opposed to oil which is used by industrial and transport and mining in a small form like and a lower end by commercial users. These kind of conclusions can be readily drawn without much effort by the audience or the analyst trying to put more context into the slide, more context into the graphic.

This is very useful to present flow data over a period of events. Another interesting use case of Sankey diagram is international trade flow data which can be used to understand how trade is changing between countries over a given events. Also within the company the flow of resources can be seen between two different parts using a Sankey diagram and these are very intuitive and are increasingly used to understand the flow data.

Now we also look at another comparison plot called small multiples. This is nothing but a bar graph or a line graph but represented again and again over a particular feature. Let me explain this using an example.

In this case we are looking at the product and sales of a company. The company has about six to seven products like it's a toy store it has six to seven products and it sells in seven eight countries. Now the traditional way to compare this would be the sales across countries would be you take a particular product category and do bar graph where the length of the bar would represent the sales in that particular country for that particular sub item.

Let us say action figures or Lego bricks. However using small multiples the same bar graph is repeated again and again in the same space. So now you can clearly compare the sales of overall sales and see which country is doing better and also see which is the largest product that is generating highest revenue in that particular country and can simultaneously compare that with other countries.

So in this way small multiples by dividing the graph based on the region of the country or any particular attribute enables comparison within that category and across categories in a simultaneous space. It is very powerful to do this comparisons if you have actually multiple units to be compared simultaneously. You would avoid clutter on your bar graph and present it in more readable format simultaneously enabling comparisons within and across the groups.

-----End-----